

TCS349: Responsible and Explainable AI

Unit 1, Lecture 3

Common Ethical Dilemmas in AI and Stakeholder-Oriented Ethical Analysis

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Warm-Up Question: When does an AI problem become an ethical problem?

Suppose an AI system is accurate but harms a minority group.

Suppose an AI tool is useful but collects too much personal data.

Suppose an autonomous system must choose between two harmful outcomes.

Question for students: Is this only a technical issue, or is it also an ethical issue?



What is an Ethical Dilemma?

An ethical dilemma is a situation in which two or more important values, duties or choices conflict.

In such situations, there may be no perfect option.

A technically correct decision may still be ethically problematic.

In AI, dilemmas often involve fairness, privacy, safety, accountability, transparency and human autonomy.

Why do Ethical Dilemmas Arise in AI?

AI systems operate at scale and can affect many people at once.

They learn from historical data, which may reflect past inequalities.

They may optimise one goal, such as accuracy or efficiency, while ignoring other values.

Different stakeholders may want different outcomes.

Real-world deployment creates trade-offs between privacy, fairness, safety, cost and convenience.

Common Ethical Dilemmas in AI

Fairness versus accuracy

Privacy versus personalisation

Safety versus autonomy

Transparency versus performance of complex models

Innovation versus regulation

Human control versus automated decision-making

Dilemma 1: Autonomous Vehicles and the Trolley Problem

An autonomous vehicle may face a situation where harm cannot be fully avoided.

Example: Should the system protect passengers first, or minimise total harm overall?

Key ethical values involved: safety, human life, accountability and public trust.

Important teaching point: even if such situations are rare, they help us think about how AI should make high-stakes decisions.



Dilemma 2: AI in Hiring and Resume Screening

Companies may use AI to shortlist job applicants.

If past hiring data was biased, the AI may repeat or amplify that bias.

A system may appear efficient but still unfairly disadvantage women, minorities or non-traditional candidates.

Ethical questions: Who is harmed? Who benefits? Who is accountable?



Dilemma 3: Surveillance, Privacy and Facial Recognition

Facial recognition may improve security and convenience.

At the same time, it can create risks of mass surveillance, profiling and misuse of personal data.

People may be monitored without meaningful consent.

The dilemma: how do we balance collective security with individual privacy and civil liberties?



Dilemma 4 and Dilemma 5: Additional Real-World Cases

Healthcare AI

AI can support diagnosis and treatment decisions.

But a wrong prediction may directly affect patient health.

Should a doctor always follow AI recommendations?

How much explanation should patients receive?

Generative AI and Deepfakes

Generative AI can create useful text, code, images and audio.

It can also produce misinformation, impersonation and harmful synthetic media.

Should open access be encouraged, or should stricter controls be applied?

How should responsibility be assigned when generated content causes harm?

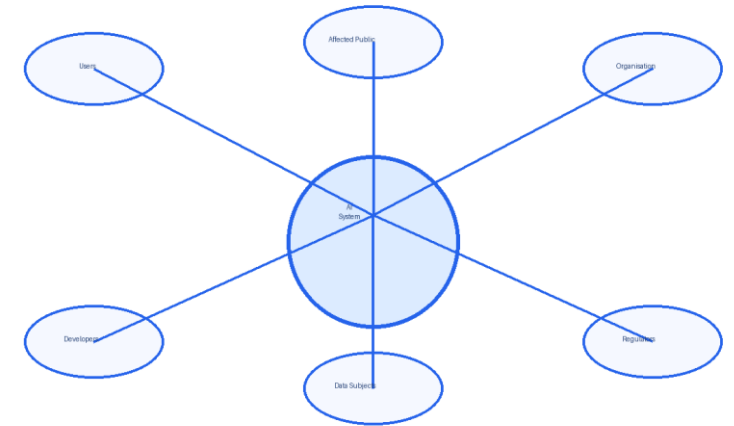
From Technical View to Stakeholder View

A purely technical view asks: Does the model work?

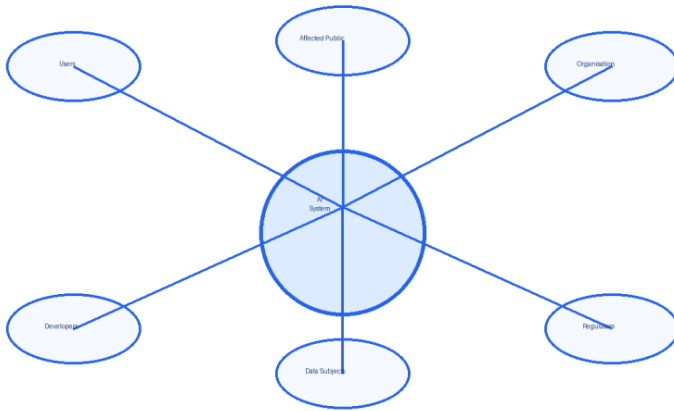
A stakeholder view asks: Who is affected, how are they affected and what values are at stake?

Stakeholder-oriented ethical analysis broadens the discussion beyond model performance.

It helps us move from ‘Can we build it?’ to ‘Should we build it, and how should we use it responsibly?’



Who are the Stakeholders in an AI System?



Developers and data scientists

The organisation deploying the AI system

Direct users of the system

Data subjects whose data is used

People affected by decisions, even if they never use the system

Regulators, policy-makers and society at large

A Simple Stakeholder-Oriented Ethical Analysis Framework

Step 1: Describe the AI system and its purpose.

Step 2: Identify the stakeholders.

Step 3: List the expected benefits.

Step 4: Identify possible harms, risks and trade-offs.

Step 5: Examine values involved: fairness, privacy, safety, accountability, autonomy and transparency.

Step 6: Suggest safeguards, design changes or policy controls.

Worked Example: Social Media Recommendation System

Possible Benefits

Personalised content

Higher user engagement

Convenience and relevance

Business growth for the platform

Possible Ethical Concerns

Echo chambers and misinformation

Addiction and excessive screen time

Privacy concerns from profiling

Unequal impact on different age groups or communities